화학특강 (2010년 2학기)
제1장 화학의 기본 개념

황현석
강원대학교 화학과

2010년 9월 9일 목요일

Units

1. Units
   1) SI prefix (metric prefix) (SI = Systemè International)

SI prefix	Symbol	Power of 10	SI prefix	Symbol	Power of 10
peta	P	10^{15}	milli	m	10^{-3}
tera	T	10^{12}	micro	μ	10^{-6}
giga	G	10^{9}	nano	n	10^{-9}
mega	M	10^{6}	pico	p	10^{-12}
kilo	k	10^{3}	femto	f	10^{-15}
centi	c	10^{-2}	atto	a	10^{-18}

Units

1. Units

   1) Volume

      - 1 mL = 1 cm^3 = 1 cc
      - 1 gallon = 3.785 L = 8 pint

   2) Mass and weight

      - 1 lb ≈ 0.453 kg

   3) Density

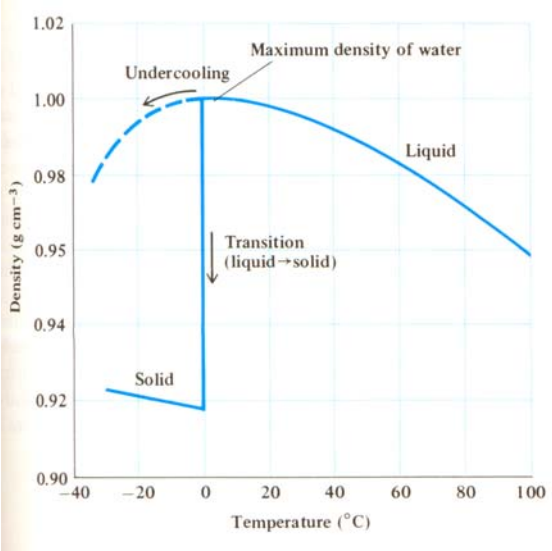
      - Density = Mass / Volume
      - Water density = 0.99707 g/cm^3 (25°C)



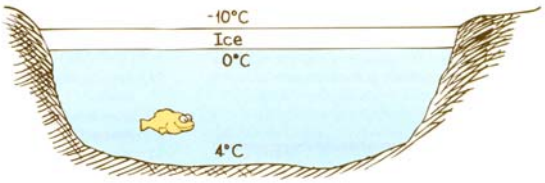
(예제 1) 25°C에서 물의 밀도를 1.00 g/cm^3라고 할 때, 물의 몰농도를 구하여라.

Water density

- Water density



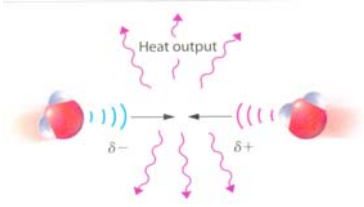
- Why do lakes freeze from the top down?



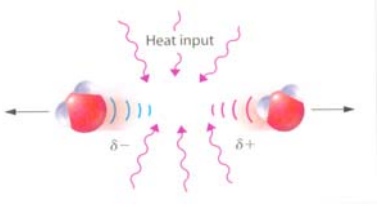
1 기압 하에서 얼음의 융해 과정에 대한 엔탈피, 엔트로피, 그리고 Gibbs 자유에너지 변화

T (K)	ΔH_{fus} (kJ mol^{-1})	ΔS_{fus} (kJ mol^{-1}K^{-1})	ΔG_{fus} (kJ mol^{-1})
263	5.619	0.0206	0.213
273	6.008	0.0220	0.0
283	6.397	0.0234	-0.226

물의 융해 과정에서 ΔH와 ΔS가 0보다 큰 이유는?

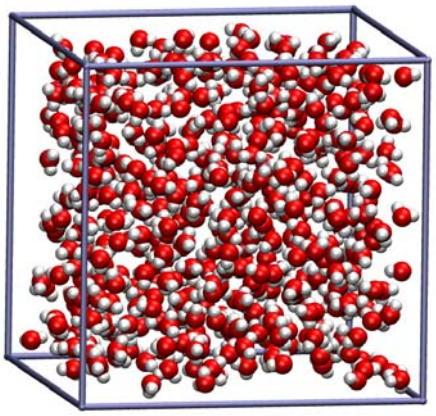


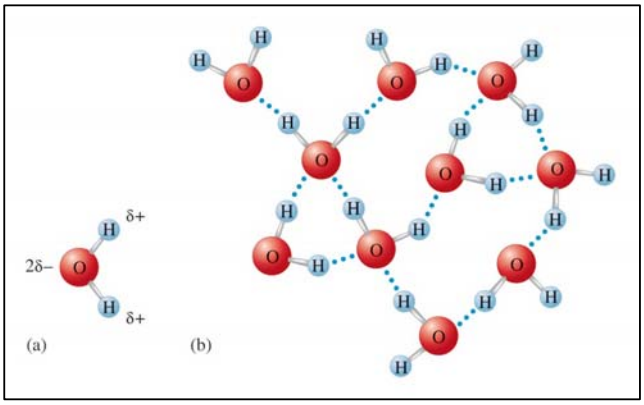
얼음이 되는 경우 열 방출 ($\Delta H < 0$)



물이 되는 경우 열 흡수 ($\Delta H > 0$)

Structure of Liquid Water

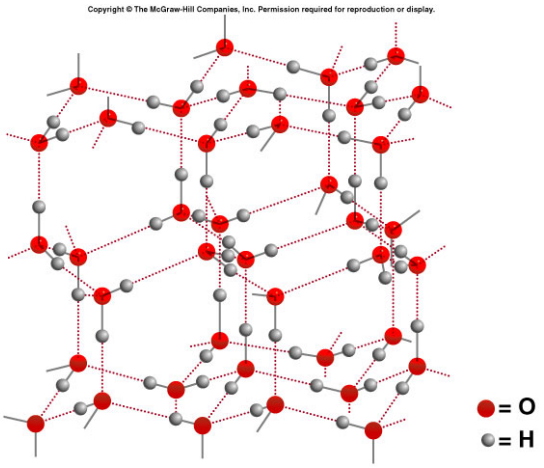




액체 상태에서 물분자들의 배열이 얼음 상태에서 물분자들의 배열보다 더 무질서하므로 액체 상태 물의 엔트로피가 더 크다. 점선은 산소원자와 수소원자 간의 수소결합을 나타낸다.

Structure of Iced Water (I)

- Three-dimensional structure of ice

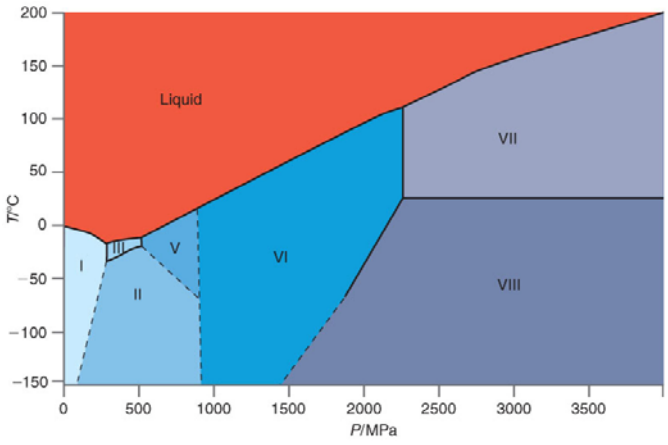

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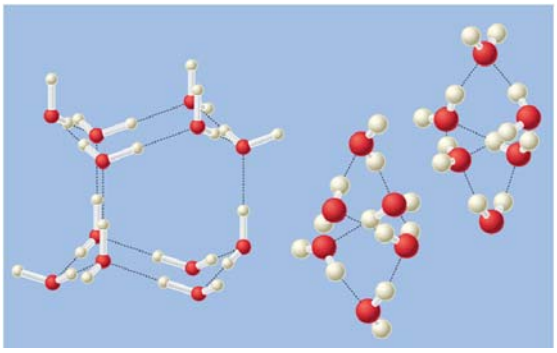


얼음의 3 차원 입체 구조. 한 개의 산소 원자는 네 개의 수소 원자와 공유 결합 혹은 수소 결합에 의해 연결되어 있다. 점선은 산소 원자와 수소 원자 간의 수소 결합을 나타낸다.

Structure of Iced Water (II)

- 온도와 압력에 따른 물의 서로 다른 고체상: 얼음의 여러가지 구조

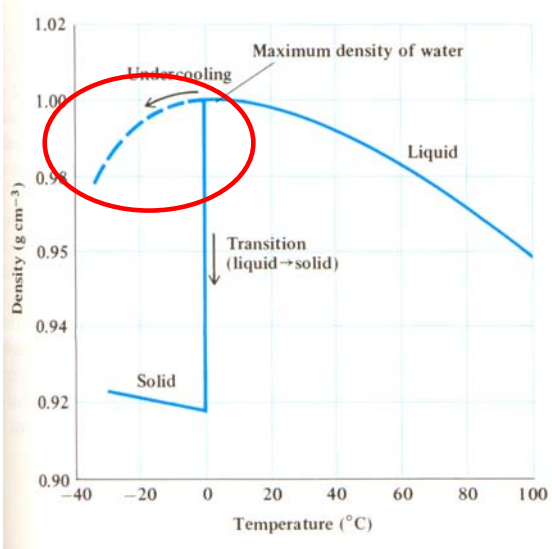




물의 서로 다른 두 가지 결정구조: 얼음 I과 얼음 VI

육방 얼음 (얼음 I)의 밀도는 0°C, 1기압에서 0.9167 g/cm^{3}이고 얼음 VI의 밀도는 1.6 GPa에서 1.31 g/cm^{3}이다. 이 압력에서 액체물의 밀도는 1.18 g/cm^{3}이다. 얼음 VI이 물에 뜰까? 가라앉을까?

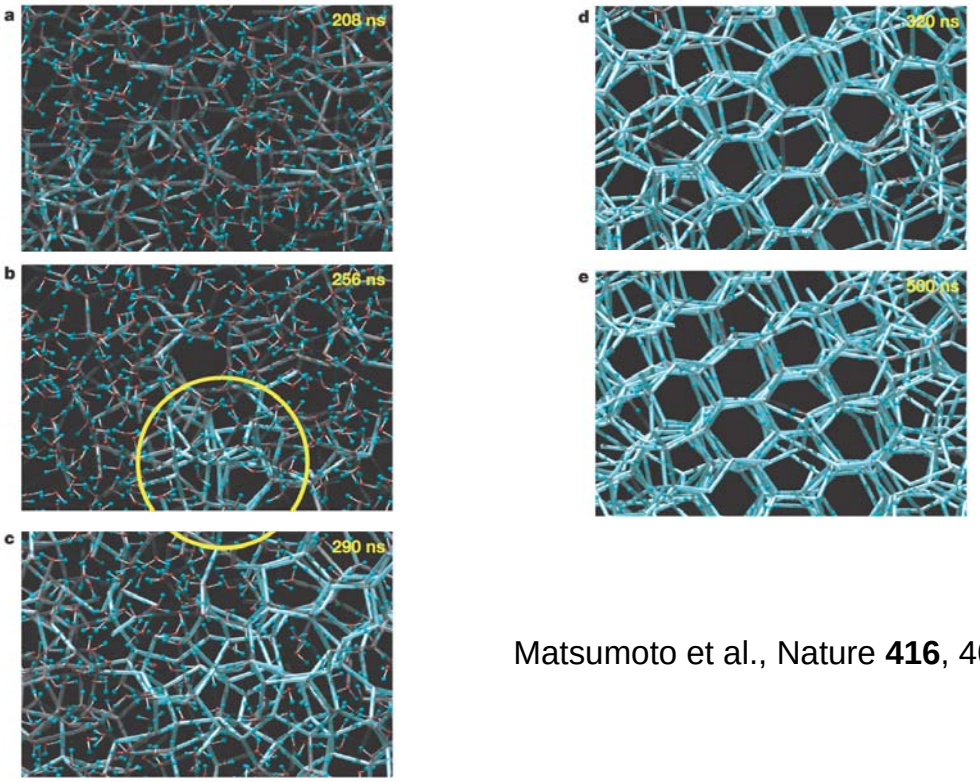
Supercooled Water





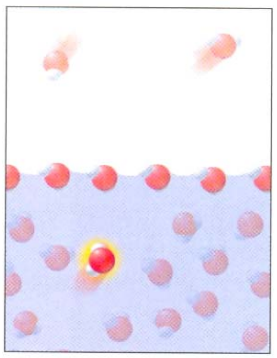


Supercooled Water: Molecular Dynamics Simulation (분자동역학 모의실험)

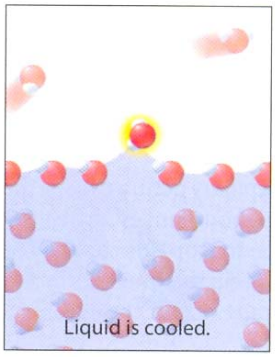


Matsumoto et al., Nature **416**, 409 (2002)

Water vaporization



수소 결합을 끊을 수 있을 정도의 운동 에너지를 가진 물 분자가 표면으로 이동



높은 운동 에너지를 가진 물 분자가 공기 중으로 날아갈 때 물의 온도는 낮아진다.



운동 에너지를 잃은 기체 상태의 물 분자를 받아들인 공기의 온도도 낮아진다.





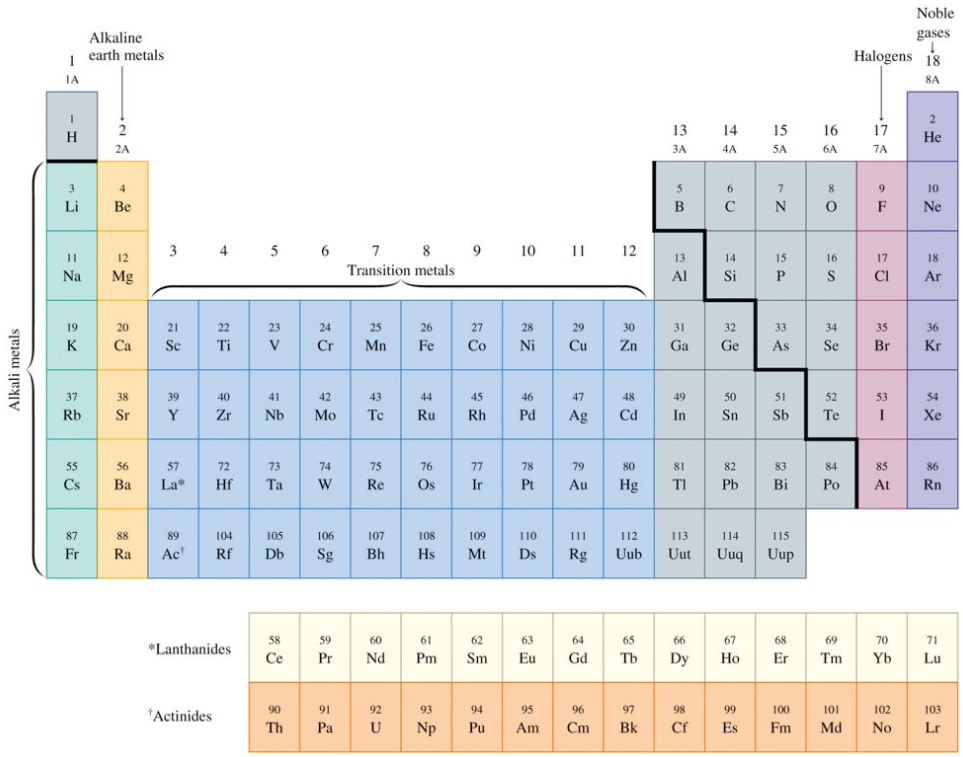
Normal boiling points and standard enthalpy changes and standard entropy changes of vaporization

Table 18.1 Normal boiling points and standard enthalpy changes and standard entropy changes of vaporization

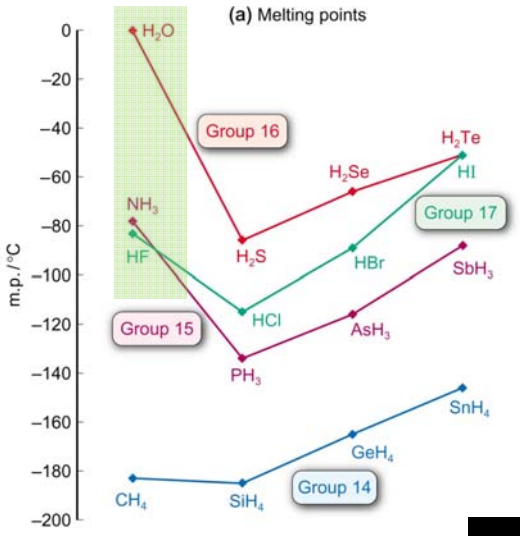
	b.p./°C	$\Delta_{vap}H^{\ominus}$/kJ mol^{-1}	$\Delta_{vap}S^{\ominus}$/J K^{-1} mol^{-1}
Benzene (C_6H_6)	80.1	+30.8	+87.2
Carbon disulfide (CS_2)	46.3	+26.7	+83.7
Hydrogen sulfide (H_2S)	−60.4	+18.7	+87.9
Tetrachloromethane (CCl_4)	76.7	+30.0	+85.8
Cyclohexane (C_6H_{12})	80.7	+30.1	+85.1
Decane ($C_{10}H_{22}$)	174	+38.8	+86.7
Methoxymethane ($(CH_3)_2O$)	−23.0	+21.5	+86.0
Methanol (CH_3OH)	65.0	+35.2	+104.1
Ethanol (C_2H_5OH)	78.3	+38.6	+110.0
Water (H_2O)	100.0	+40.7	+109.1
Butane (C_4H_{10})	−0.50	+22.4	+82.1
Methylbenzene ($C_6H_5CH_3$)	110.7	+33.5	+87.2
Ethanoic acid (CH_3COOH)	118.3	+24.4	+61.9

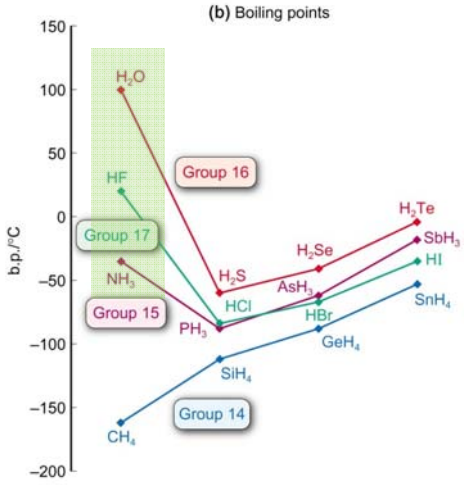
Trouton's rule:
$\Delta S_{vap}^{\circ} \approx +85$ J K^{-1} mol^{-1}

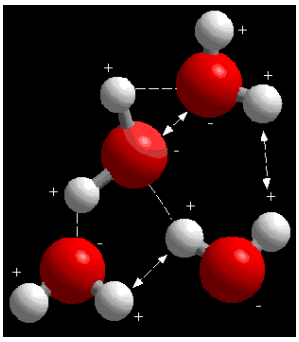
Periodic Table



The Hydrogen Bond







Origin of the Hydrogen Bond

1. Large electronegativity difference between H and F, O, and N.
   - Electronegativity: the ability of an atom to attract toward itself the electrons in a chemical bond.
   - Linus Pauling

$$\chi_A - \chi_B = (eV)^{-1/2}\sqrt{E_d(AB) - [E_d(AA) + E_d(BB)]/2}$$

Electronegativity of HBr = 0.73
(dissociation energies:
H–Br, 3.79 eV; H–H, 4.52 eV; Br–Br 2.00 eV)

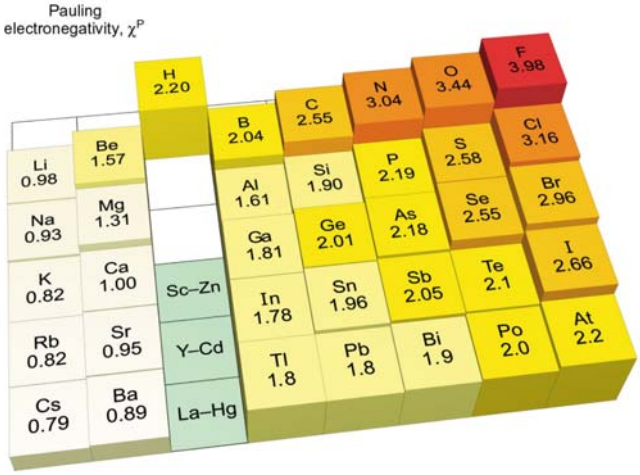


TABLE 2.2 Bond Dissociation Energies of Diatomic Molecules (kJ·mol^{-1})

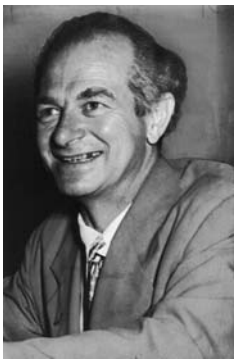
Molecule	Bond dissociation energy
H_2	424
N_2	932
O_2	484
CO	1062
F_2	146
Cl_2	230
Br_2	181
I_2	139
HF	543
HCl	419
HBr	354
HI	287

1 eV = 96.485 kJ mol^{-1}

Chang, 8.3, 9.5, 9.10, and 11.2



Marie Skłodowska Curie
Physics in 1903
Chemistry in 1911



Linus Pauling
Chemisty in 1954
Peac Prize in 1962



John Bardeen
Physics in 1956
Physics in 1972

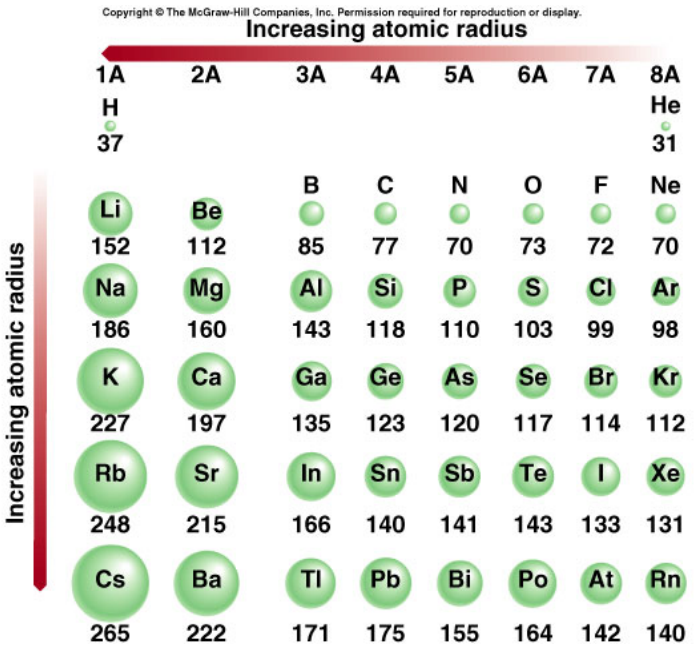


Frederic Sanger
Chemistry in 1958
Chemistry in 1980

Origin of the Hydrogen Bond

2. Small size of F, O, and N.

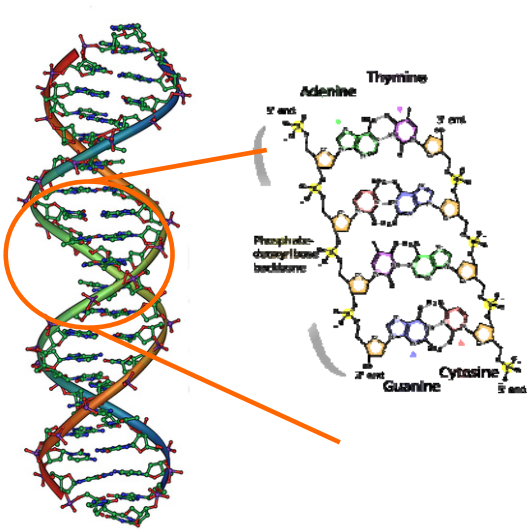
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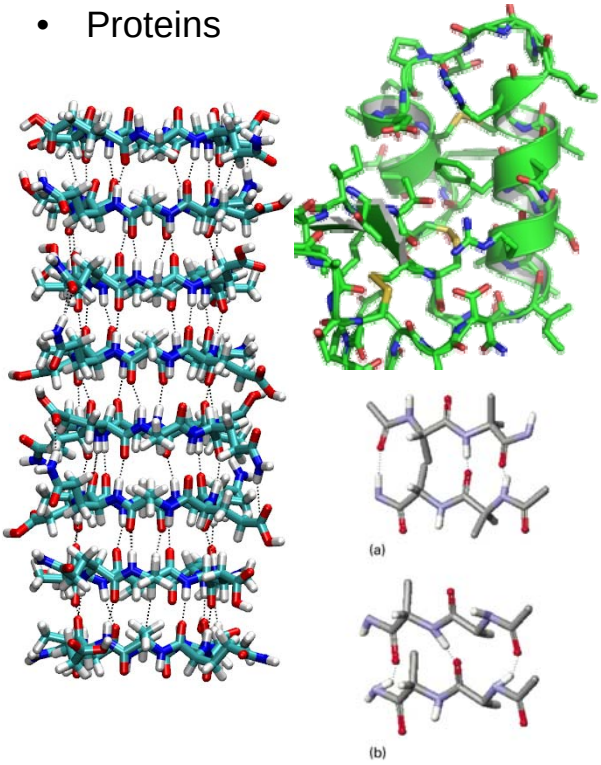
Chang, 8.3, 9.5, 9.10, and 11.2

The Hydrogen Bond in Biological Systems

- DNA double helix



- Proteins

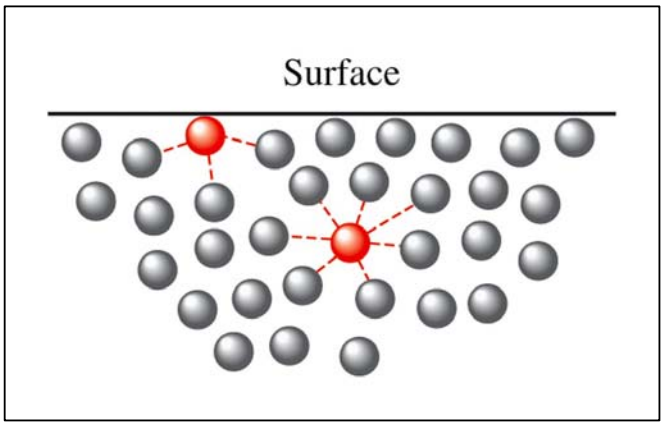


Water Hydrogen Bond and Surface Tension

- Surface tension





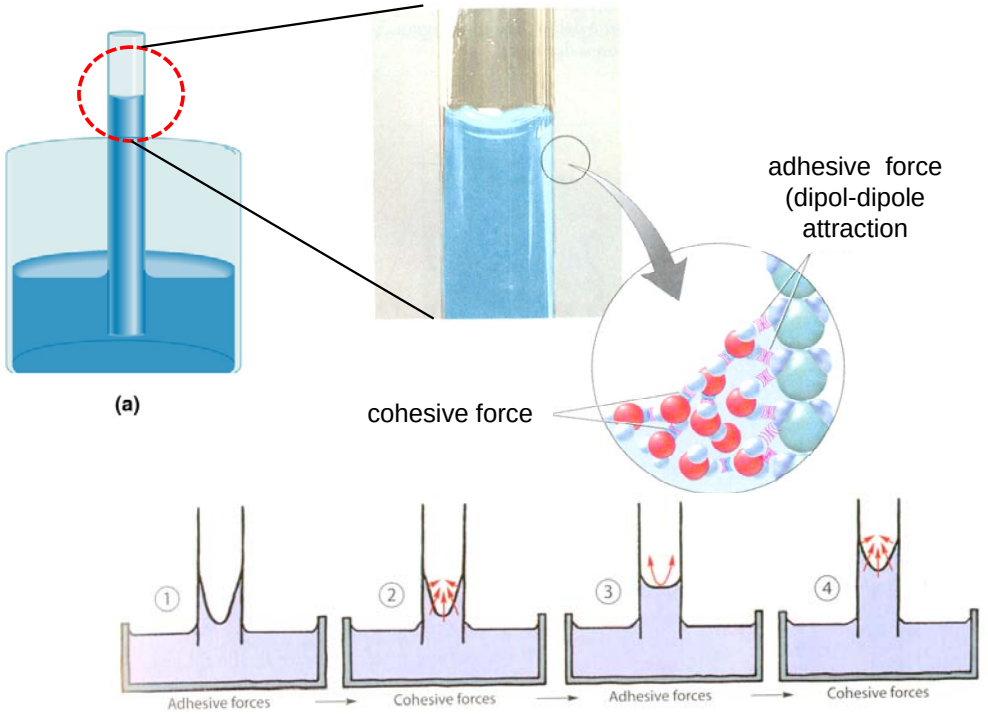


액체 표면에 있는 분자들은 내부에 있는 분자들보다 끌어당김 상호작용 에너지가 더 작고 그 결과 더 불안정해진다.

Chang, 11.3

Cohesion and adhesion

- cohesion and adhesion
  - cohesion: the intermolecular attraction between like molecules.
  - adhesion: an attraction between unlike molecules.



Chang, 11.3

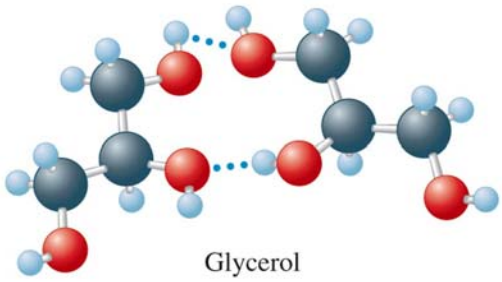
Viscosity

점성도 (viscosity): 흐름에 대한 유체 저항의 척도.

물의 수소결합과 글리세롤의 수소결합은 점성도를 증가시킨다.

TABLE 11.3 Viscosity of Some Common Liquids at 20°C

Liquid	Viscosity (N s/m^2)*
Acetone (C_3H_6O)	3.16×10^{-4}
Benzene (C_6H_6)	6.25×10^{-4}
Blood	4×10^{-3}
Carbon tetrachloride (CCl_4)	9.69×10^{-4}
Ethanol (C_2H_5OH)	1.20×10^{-3}
Diethyl ether ($C_2H_5OC_2H_5$)	2.33×10^{-4}
Glycerol ($C_3H_8O_3$)	1.49
Mercury (Hg)	1.55×10^{-3}
Water (H_2O)	1.01×10^{-3}







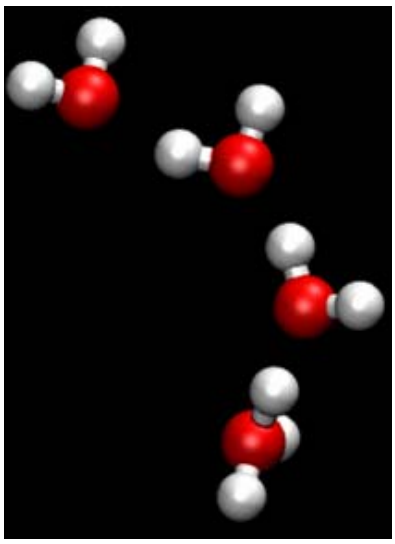
Ionic conductivity

Table 17.1 Limiting ionic conductivities, λ, in water / 10^{-3} S m^2 mol^{-1}

Cations		Anions	
H^+ (H_3O^+)	35.0	OH^-	19.9
Li^+	3.9	F^-	5.5
Na^+	5.0	Cl^-	7.6
K^+	7.4	Br^-	7.8
Cs^+	7.7	I^-	7.7
Ag^+	6.2	HCO_2^-	5.5
NH_4^+	7.4	NO_3^{2-}	7.2
Mg^{2+}	10.6	SO_4^{2-}	13.9
Ca^{2+}	11.9	CO_3^{2-}	17.0
Sr^{2+}	11.9	$CH_3CO_2^-$	4.1

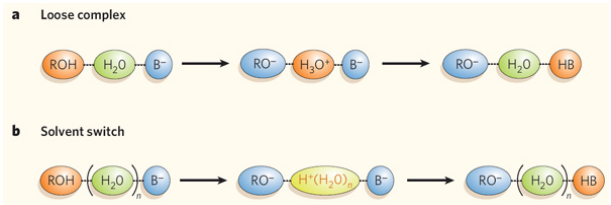
Grotthus Mechanism and Proton Transfer

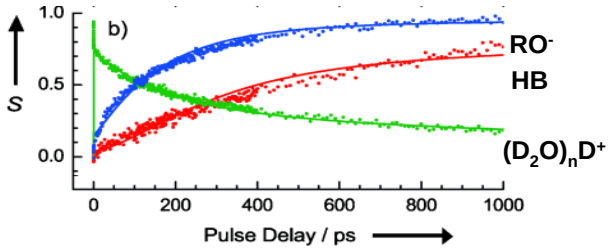
- Grotthus mechanism



Marx, ChemPhysChem **7**, 1849 (2006)

- Proton transfer in acid and base
  - Arrhenius definition
  - Brønsted definition
  - Lewis defintion





Mohammed et al., Angew. Chem. Int. Ed. **46**, 1458 (2007)

Chang, 15.1

Diffusion Coefficients

- Diffusion: the event where two or more gas or liquid molecules come into contact and mix.

Synoptic table 21.8* Diffusion coefficients at 298 K

	$D/(10^{-9}\ m^2\ s^{-1})$
H^+ in water	9.31
I_2 in hexane	4.05
Na^+ in water	1.33
Sucrose in water	0.522

* More values are given in the *Data section*.

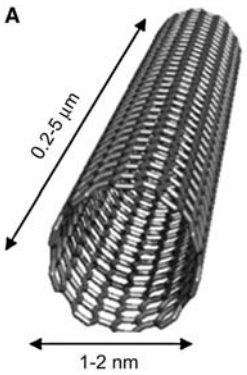
- Relation between ionic conductivity (λ) and diffusion coefficient (D)

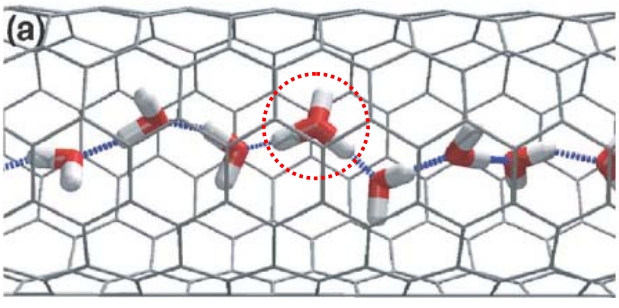
$$\lambda = \frac{z^2 D F^2}{RT}$$

Aktins' Physical Chemistry, Chapt. 21

Proton diffusion in carbon nanotubes

- Carbon nanotube



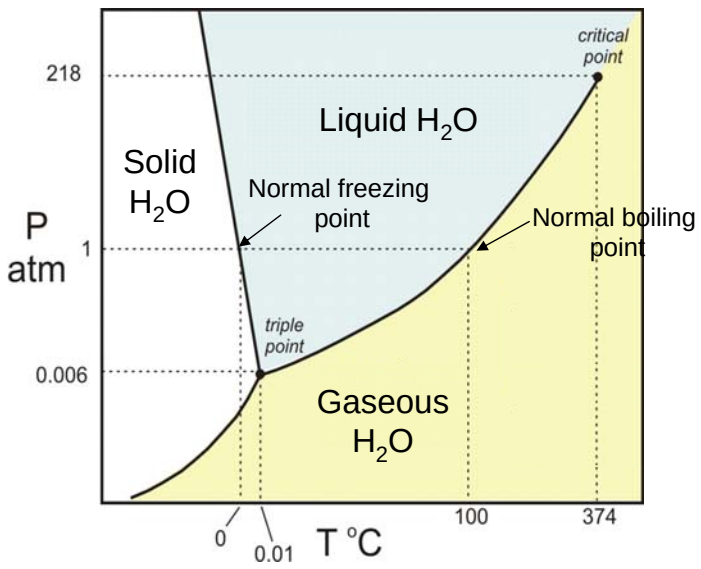


system	D_{H+} (Å²/ps)
Bulk water (simulation)	0.4
Water in carbon nanotube (simulation	3-17
Bulk water (experiment)	0.93

Dellago et al., Phys. Rev. Lett. **90**, 105902 (2003)

상평형도 (Phase diagram)

- Water phase diagram



Chang, General Chemistry 8th ed., Chapter 11

- 삼중점 (triple point): 세 상태가 서로 평형을 유지하는 유일한 상태.
- 임계온도 (critical temperature, T_c): 이 온도 이상에서는 아무리 높은 압력을 가하더라도 기체가 액화될 수 없는 온도.
- 임계압력 (critical pressure, P_c): 임계온도에서 액화시키기 위해 가해주는 최소 압력
- 임계점 (critical point): 상평형도에서 임계온도와 임계압력에 의해 정의되는 상태.
- 초임계유체 (supercritical fluid): 임계점을 지나 존재하는 물질의 상태.

화학 퍼텐셜 (chemical potential)

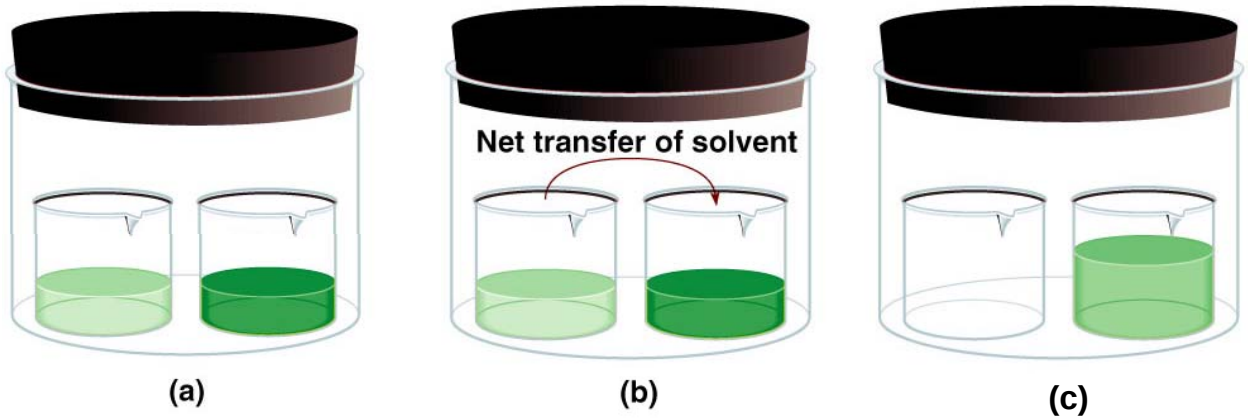
- 화학 퍼텐셜: 일정한 상에 있는 물질이 다른 상으로 움직이려는 경향.
- 물질은 화학 퍼텐셜이 높은 상에서 낮은 상으로 움직이려는 경향이 있다.
- 이상기체의 화학 퍼텐셜

$$\mu(T,P) = \mu^{\circ}(T) + RT\ln\left(\frac{P}{P^{\circ}}\right)$$

- 이상액체의 화학 퍼텐셜

$$\mu_{용매}(T,P) = \mu^{*}_{용매}(T,P) + RT\ln x_{용매}$$

화학퍼텐셜



Chang, 12.6

Clapeyron and Clasius-Clapeyron Equations

- Clapeyron equation

$$\frac{dP}{dT} = \frac{\Delta_{fus}H}{T\Delta_{fus}V}$$

- Clausius-Clapeyron equation

$$\frac{dP}{dT} = \frac{\Delta_{vap}H}{T\Delta_{vap}V}$$

$$P \approx P^* + \frac{\Delta_{fus}H}{\Delta_{fus}V}(T - T^*)$$

$$\ln\left(\frac{P}{P^*}\right) \approx -\frac{\Delta_{fus}H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right)$$